Nonuniqueness of a symmetric word equation in two-by-two positive definite matrices

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Solved The order-two uniqueness assertion is disproved by real positive definite counterexamples. The dated independent agent review records the subsequent checks and supersedes original draft remarks about pending review. This is not external human peer review or formal certification.

Abstract

The symmetric word equation $XBX^{12}BX = P$, with $B = \left[\begin{smallmatrix} 1 & 4 \\ 4 & 17 \end{smallmatrix}\right]$ and an explicit integer matrix P, has at least three distinct real symmetric positive definite solutions. At $X_0 = \operatorname{diag}(3, 1)$ the Jacobian determinant in symmetric-matrix coordinates is exactly -11785057051824. Armstrong–Hillar's degree theorem therefore already proves nonuniqueness, disproving their Conjecture 11.5. Separate integer interval-arithmetic certificates establish two further roots without using the degree theorem. A factored Jacobian also gives a sharp classification within the family XBX^rBX: universal uniqueness holds for integer $1 \le r \le 6$, whereas counterexamples exist for every integer $r \ge 7$.

1 An explicit counterexample

Write $\mathcal{P}_2$ for the open cone of real symmetric positive definite 2×2 matrices. Coordinates on the space of symmetric matrices are always (X_{11}, X_{12}, X_{22}), with the same order in the domain and codomain. For fixed $B \in \mathcal{P}_2$, put $F_r(X) = XBX^rBX$.

Theorem 1. *Let*

$$B = \begin{pmatrix} 1 & 4 \\ 4 & 17 \end{pmatrix}, \qquad X_0 = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \qquad P = \begin{pmatrix} 4783113 & 6377496 \\ 6377496 & 8503345 \end{pmatrix}.$$

Then $F_{12}(X_0) = P$ and the equation $F_{12}(X) = P$ has more than one solution in $\mathcal{P}_2$.

Proof. The leading principal minors of B are $1, 1$, and those of X_0 are $3, 3$. Direct multiplication gives the displayed P; in particular, $P_{11} > 0$ and $\det P = 3^{14} > 0$. The word $XBX^{12}BX$ is an interlaced symmetric word in the sense of Armstrong and Hillar [1].

For a symmetric perturbation H, the product rule gives

$$\begin{aligned} DF_r(X)[H] &= HBX^rBX + XBX^rBH \\ &\quad + XB\left(\sum_{j=0}^{r-1} X^j H X^{r-1-j}\right) BX. \end{aligned} \tag{1}$$

Applying (1) to the three coordinate basis matrices yields

$$JF_{12}(X_0) = \begin{pmatrix} 22320618 & 31886832 & 1728 \\ 27635000 & 36403996 & 6379944 \\ 34012224 & 40389584 & 17010158 \end{pmatrix},$$

whose determinant is

$$\det JF_{12}(X_0) = -11785057051824 < 0. \tag{2}$$

Corollary 11.1 of [1] states that a negative Jacobian determinant at a positive definite point of an interlaced symmetric word map implies another real positive definite solution with the same image. It applies directly to (2). □

Theorem 1 is a counterexample to Conjecture 11.5 of [1], which asserts uniqueness of symmetric word equations in dimension two. This is not the previously studied word $XBX^2B^3X^2BX$, whose dimension-two uniqueness is established in [1, 3]. The word and the dimension are both essential.

2 A factored Jacobian and a sharp family threshold

Lemma 2. *Let $X = \operatorname{diag}(t, 1)$, $t > 0$, and $B = \left[\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right] \in \mathcal{P}_2$. For an integer $r \geq 1$ put $p = t^r$ and $g = \sum_{j=0}^{r-1} t^j$. Then*

$$\det JF_r(X) = t^{r+1}(r+2)(ac-b^2)^2 H_r, \tag{3}$$

where

$$H_r = (r+2)(a^2t^{r+1} + c^2 + acgt) - (r-2)b^2(gt + t^r + t). \tag{4}$$

Proof. Formula (1) gives the Jacobian

$$\begin{pmatrix} (r+2)a^2pt + 2b^2t & 2bt(agt + ap + c) & rb^2t^2 \\ (r+1)abp + bc & a^2pt + c^2 + acgt + b^2(gt + p + t) & bt(ap + (r+1)c) \\ rb^2p/t & 2b(ap + c(g+1)) & (r+2)c^2 + 2b^2p \end{pmatrix}.$$

Expanding its determinant and collecting terms gives (3)–(4). This identity remains valid at $t = 1$, since g is a polynomial rather than a quotient. □

For the data of Theorem 1, formula (4) gives $H_{12} = -527992$ and hence

$$\det JF_{12}(X_0) = 3^{13} \cdot 14 \cdot (-527992),$$

an additional short check on the integer determinant.

Theorem 3. *For every integer $1 \leq r \leq 6$ and every $B, P \in \mathcal{P}_2$, the equation $XBX^rBX = P$ has exactly one solution in $\mathcal{P}_2$. For each integer $r \geq 7$, some $B, P \in \mathcal{P}_2$ give more than one solution.*

Proof. An orthogonal change of basis diagonalizes X, and a positive scalar multiple puts it in the form $\operatorname{diag}(t, 1)$. The change of coordinates acts identically on domain and codomain, so preserves the sign of the Jacobian determinant. Positive scaling also preserves its sign by homogeneity.

For $r = 1, 2$, (4) is positive. For $3 \leq r \leq 6$, use $b^2 < ac$ and $r - 2 > 0$ to obtain

$$H_r > (r+2)(a^2t^{r+1} + c^2) + ac\{4gt - (r-2)(t^r + t)\},$$

$$4gt - (r-2)(t^r + t) = (6-r)(t + t^r) + 4\sum_{j=2}^{r-1} t^j \geq 0.$$

Thus the Jacobian determinant is positive everywhere on $\mathcal{P}_2$. The degree-one theorem for interlaced symmetric word maps, Theorem 1.5 of [1], now implies existence and uniqueness: every value is regular, every inverse image contributes $+1$, and the total degree is one.

Conversely fix $r \geq 7$ and choose

$$0 < \varepsilon < \frac{r-6}{2(r-2)}, \quad c = t^{(r+1)/2}, \quad a = 1, \quad b^2 = (1-\varepsilon)c.$$

These choices give $\det B = \varepsilon c > 0$. From (4),

$$\frac{H_r}{ct^r} = (r+2)\left(2t^{(1-r)/2} + \frac{gt}{t^r}\right) - (r-2)(1-\varepsilon)\left(\frac{gt}{t^r} + 1 + t^{1-r}\right).$$

Since $gt/t^r \to 1$ as $t \to \infty$, the limit is

$$6 - r + 2(r-2)\varepsilon < 0.$$

Therefore $H_r < 0$ for sufficiently large t. Lemma 2 and Corollary 11.1 of [1] give nonuniqueness for $P = F_r(\operatorname{diag}(t,1))$. □

3 Independent certification of three solutions

The following computer-assisted strengthening is not needed for the preceding analytic counterexample. It supplies explicit enclosures and an independent existence proof.

Theorem 4. *The matrices B, P in Theorem 1 admit at least three distinct solutions in $\mathcal{P}_2$ to $XBX^{12}BX = P$.*

Certificate proof. Since $\det B = 1$ and $\det P = 3^{14}$, every positive definite solution has determinant 3. Parameterize this surface by

$$X(x,y) = \begin{pmatrix} x & y \\ y & (3+y^2)/x \end{pmatrix}, \qquad x > 0.$$

Define the two-component function

$$f(x,y) = \big((F_{12}(X(x,y)) - P)_{11}, (F_{12}(X(x,y)) - P)_{12}\big).$$

A zero of f with $x > 0$ solves the full matrix equation: the determinants of $F_{12}(X)$ and P both equal 3^{14}, and their $(1,1)$ and $(1,2)$ entries agree, with the $(1,1)$ entry positive. Thus their remaining diagonal entries agree as well.

Two disjoint boxes of radius 10^{-40} in the (x,y) maximum norm are specified, with their full decimal centers, in the accompanying file `word_equation_interval_certificate.json`. For orientation only, their centers rounded to fourteen decimal places are

$$(x_1, y_1) \simeq (2.97299380819299, \quad 0.01891127365428),$$
$$(x_2, y_2) \simeq (3.49434510663918, -0.33048510081377).$$

The verifier uses integer fixed-point interval arithmetic with 110 decimal places and outward rounding at every operation. For each box $Q = c + [-\delta,\delta]^2$, it computes a fixed invertible rational preconditioner C from the midpoint of an interval enclosure of $Df(c)$, and verifies

$$c - Cf(c) + (I - CDf(Q))[-\delta,\delta]^2 \subset \operatorname{int} Q,$$
$$\sup_{u \in Q} \|I - CDf(u)\|_\infty < 10^{-20}.$$

Here every displayed set expression is replaced by a rigorous interval enclosure. By the mean-value theorem, $T(u) = u - Cf(u)$ maps Q into itself and is a strict contraction. The contraction mapping theorem gives a unique zero of f in each box. Both boxes have $x > 0$, are disjoint from each other, and exclude $(3,0)$. Together with X_0, they certify three distinct positive definite solutions. □

4 Reproducibility and scope

The supplied standard-library Python verifier `word_equation_certificate.py` differentiates the original product one letter at a time, checks positive definiteness by principal minors, and computes (2) by exact integer arithmetic. It does not use Lemma 2 to construct the Jacobian. The separate file `word_equation_interval_certificate.py` regenerates the two interval certificates using no floating-point arithmetic or external numerical library. The interval proof does not assume the degree theorem. An independent verifier, `word_equation_interval_independent.py`, repeats the existence proof with exact rational endpoints rather than fixed-point arithmetic and uses the Cayley–Hamilton recurrence $X^{k+1} = (\operatorname{tr} X)X^k - 3X^{k-1}$ instead of the original product evaluation. It obtains the same two strict root-box inclusions. This verification requires only Python 3; its output is recorded separately.

The counterexample resolves the dimension-two universal uniqueness conjecture, not the existence theorem or the uniqueness of every individual word. The exponent threshold in Theorem 3 is specific to the family XBX^rBX; it is not a claim about the shortest counterexample among all symmetric words. The results are submitted as complete solution claims pending external mathematical review.

References

[1] S. N. Armstrong and C. J. Hillar, *Solvability of symmetric word equations in positive definite letters*, J. London Math. Soc. **76** (2007), 777–796. doi:10.1112/jlms/jdm070. arXiv:math/0507306v4. Theorem 1.5, Corollary 11.1, and Conjecture 11.5; arXiv manuscript pp. 3, 18, and 20.

[2] C. J. Hillar and C. R. Johnson, *Symmetric word equations in two positive definite letters*, Proc. Amer. Math. Soc. **132** (2004), 945–953. doi:10.1090/S0002-9939-03-07163-6.

[3] J. Lawson and Y. Lim, *On the symmetric matrix word equation $XBX^2B^3X^2BX = A$*, Linear Algebra Appl. **427** (2007), 77–86. doi:10.1016/j.laa.2007.06.021.